

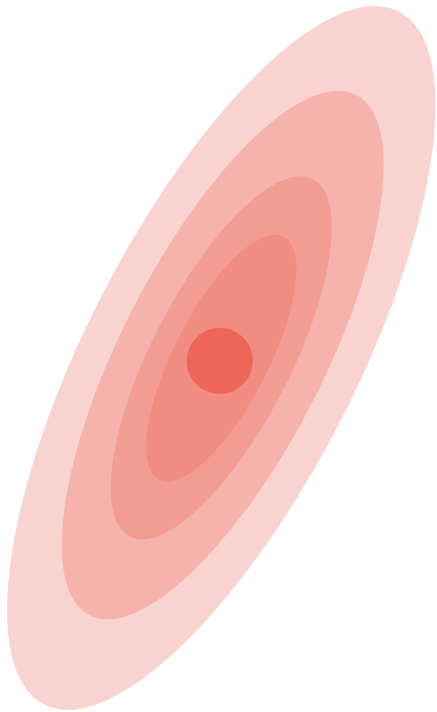
min-cost (soft)
matching

shortest path
(geodesic) in probability space

α



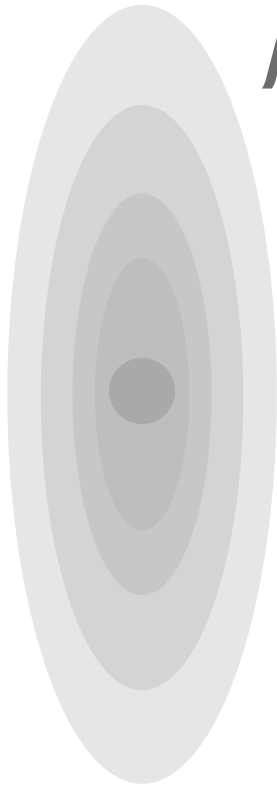
β



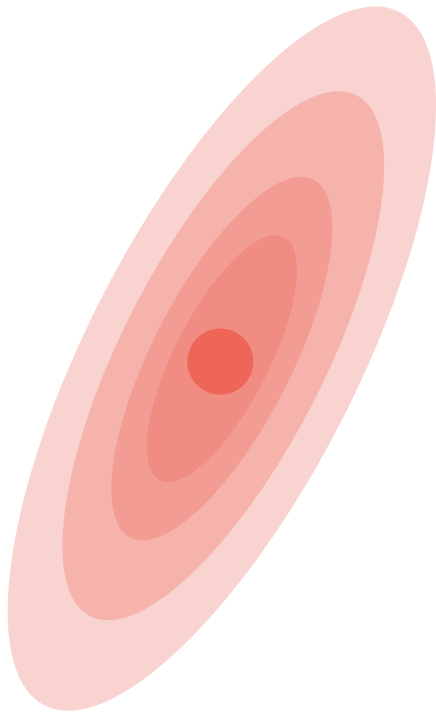
α



ρ_t



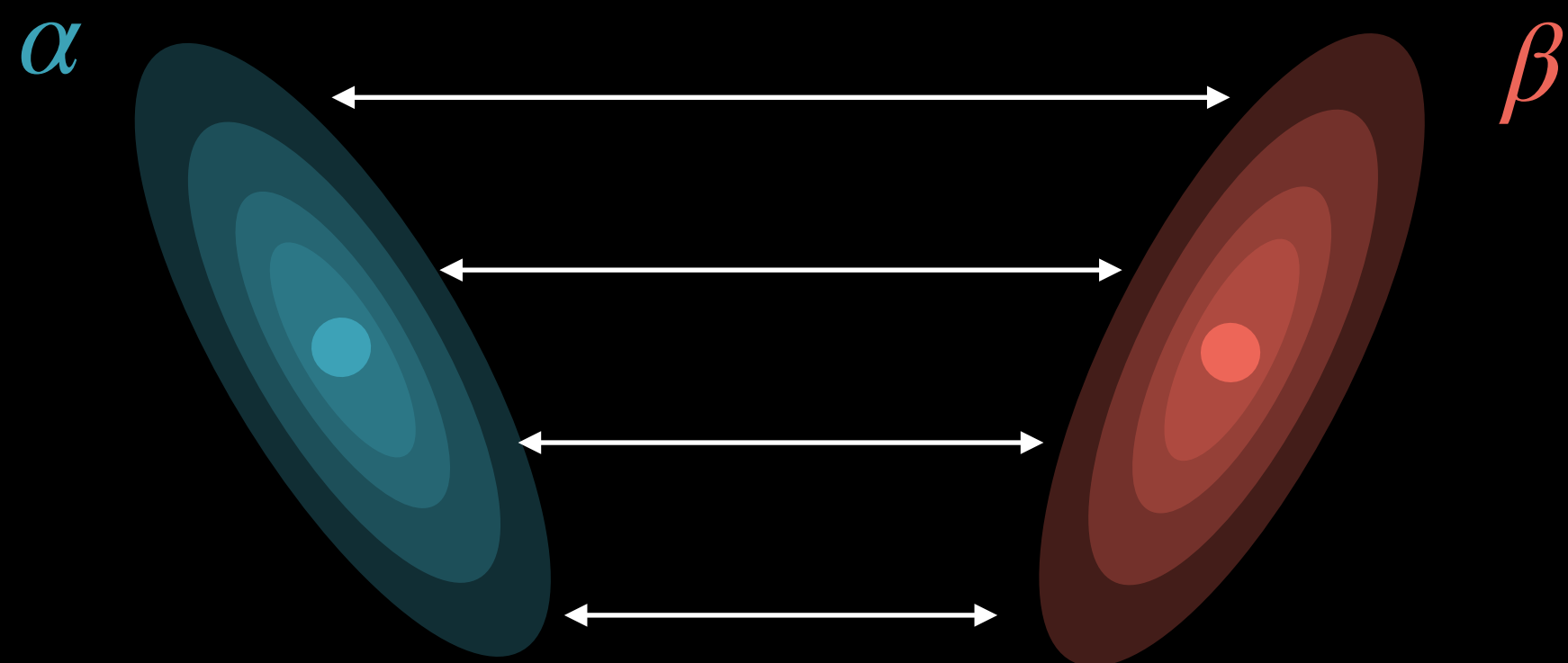
β



Static vs Dynamic OT

Static (Kantorovich)

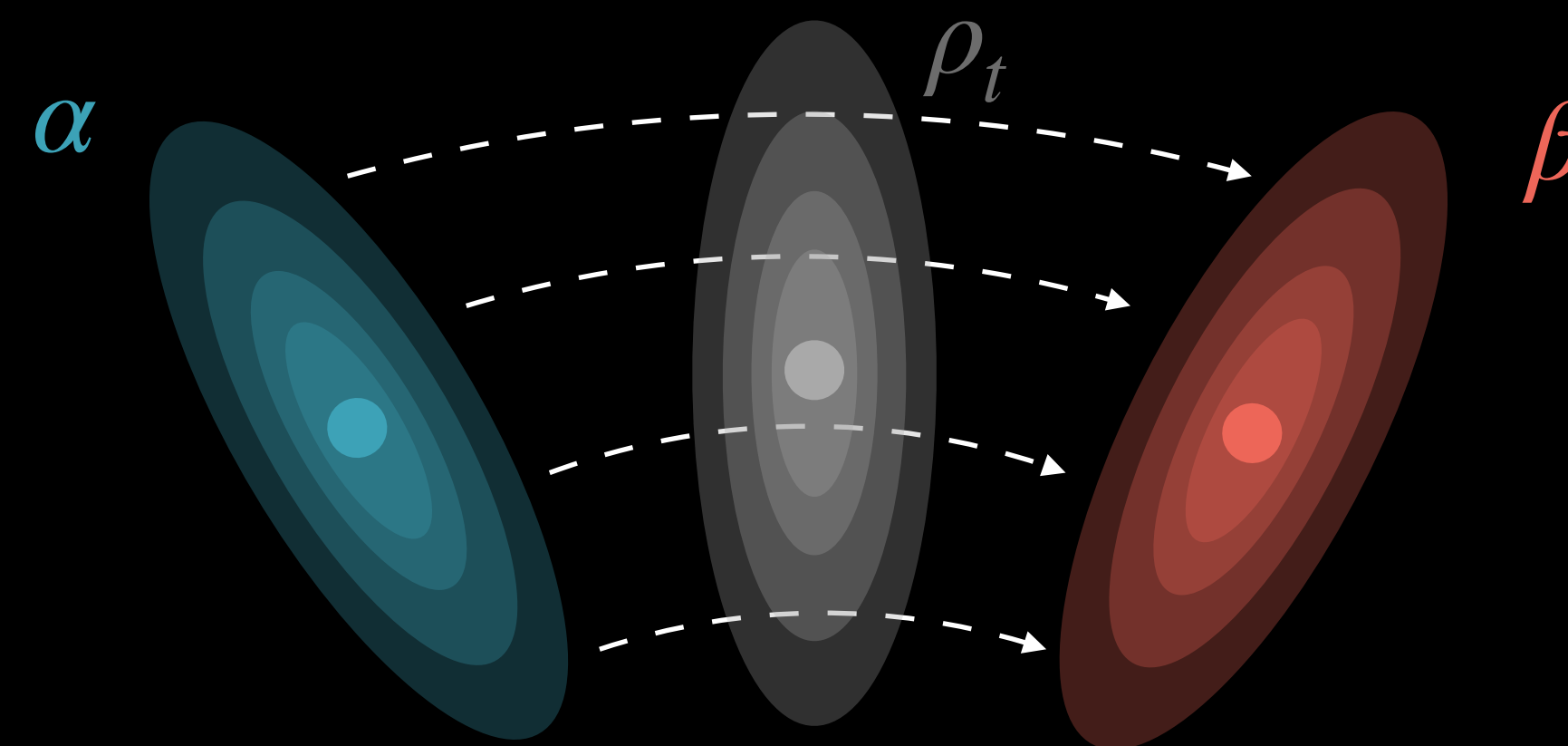
- Considers couplings transporting $\alpha \rightarrow \beta$
- Transport is one-shot, static



- Find coupling minimizing distance
- Key idea: OT as **min-cost (soft) matching** between distributions

Dynamic

- Considers flows ρ_t, v_t transporting $\alpha \rightarrow \beta$
- Transport is time-dependent, dynamic



- Find flow minimizing energy
- Key idea: OT as **shortest path (geodesic) in probability space**

Historical Context

- Benamou & Brenier (2000): equivalence between static OT and PDE formulation
- Connected OT to fluid mechanics and convex optimization
- Catalyzed methods in PDEs, physics, and ML



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A computational fluid mechanics solution to the Monge-Kantorovich mass transfer problem

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Summary. The L^2 Monge-Kantorovich mass transfer problem [31] is reset in a fluid mechanics framework and numerically solved by an augmented Lagrangian method.